Credit Risk Portfolio Analytics — Solution Overview

Executive Summary: Streamlit Views, AI Features, Model Metrics, Portfolio Insights & Data Limitations

Document: docs/solution_overview.pdf
Scope: 10k Customers | 14.7k Loan Accounts

1. STREAMLIT DASHBOARD VIEWS & TWO EMBEDDED AI FEATURES

- **Dashboard Views (4 Tabs + Global Sidebar Filters):** *Tab 1 (Portfolio Overview):* KPI cards, default rate profile across income/employment/region/age/loan type, 2D heatmaps, and cohort trends. *Tab 2 (Account & Model):* Single-account PD score, top-10 local SHAP waterfall chart, monthly EMI/DPD trajectory sparkline, and comparative model metrics. *Tab 3 (Ask the Portfolio):* Natural-language Q&A assistant translating user questions into read-only DuckDB SQL with schema guardrails. *Tab 4 (Anomaly Queue):* Filterable table of accounts flagged by peer Z-scores and IsolationForest, featuring an interactive analyst disposition logger (`Confirm/Dismiss`).
- **AI Feature 1 — "Ask the Portfolio" (LLM Q&A):** Triggered on-demand via text input; converts questions to SQL, queries the master dataframe locally, and synthesizes a plain-English answer citing source tables. Includes an expandable HITL verification box showing executed SQL and result data.
- **AI Feature 2 — Peer & Behavioral Anomaly Detection:** Triggered during ingestion; flags silent credit deterioration (high bureau utilization with 0 internal DPD) and sudden repayment shocks. Analysts review each case and log audit notes to `anomaly_reviews.json`.

2. TOP 3 PORTFOLIO INSIGHTS & DATA LIMITATIONS

- **Insight 1 (Cross-Dimensional Cut):** Low-income (`Q1/Q2`) borrowers with unsecured personal loans exhibit disproportionately high default rates when `loan_to_income_ratio` exceeds portfolio medians. *So What:* Enforce strict LTI caps on unsecured products in low-income tiers.
- **Insight 2 (External Bureau Contagion):** Business Owners and variable-income borrowers with external bureau utilization >70% default at >2x the rate of low-utilization peers. *So What:* Trigger automated quarterly bureau reviews and freeze credit lines upon cross-bureau stress.
- **Insight 3 (Early Repayment Slippage):** Accounts showing rising 3-month DPD trends and thin external bureau history show rapid roll-rates into severe default. *So What:* Implement early-stage digital cures before accounts reach 30 DPD.
- **Data Quality & Synthetic Limitations:** Because the underlying dataset is synthetically generated, `repayment_history` timestamps and periodic snapshots do not always map cleanly to individual loan origination dates, and `target_default` contains inherent random Bernoulli noise. This structural decoupling limits out-of-sample model accuracy (AUC ~0.67), reflecting a realistic noise ceiling rather than feature failure.

3. DEFAULT MODEL RESULTS & PRODUCTION READINESS

Model (80/20 Stratified Split; Collections Excluded)	AUC	Gini	KS	Evaluation & IT Ask
Logistic Regression (Train)	0.7008	0.4016	0.3065	**Metrics & Monitoring:** LightGBM and LogReg achieve stable Test AUC (~0.67) with minimal overfitting. Post-deployment monitoring requires monthly PSI, quarterly OOT AUC/KS, and A/E calibration. **Production IT Requirements:** Automated daily Airflow/CDC feeds, VPC-hosted private LLM endpoint, RBAC via SSO, PostgreSQL audit log, and MRM (SR 11-7) sign-off.
Logistic Regression (Test)	**0.6667**	**0.3334**	**0.2394**	
LightGBM (Train)	0.7222	0.4444	0.3275	
LightGBM (Test)	**0.6732**	**0.3465**	**0.2594**	